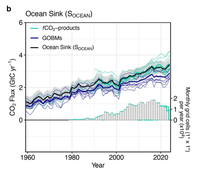
Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO_2 Reconstruction Skill in Climate Model Testbeds

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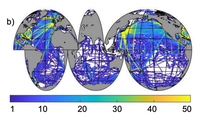
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① Introduction



Global Carbon Budget 2025
Friedlingstein et al. (2026)

- The ocean sink absorbs 29% of anthropogenic CO_2, the strongest carbon sink on Earth
- Independent pCO_2 observations are sparse → observation-based data products reconstruct the full field → but with different skills.



Number of months with 1° x 1° surface ocean fCO_2 over 1970 – 2023 (SOCATv2024)
Bakker et al. (2024)

- Hindcast & forecast models also generate the full field with coupled climate parameters → but are imperfect.
- Testbed experiments use these models as the "truth" to benchmark data products' skills.



Since models are imperfect, is there any evidence to eliminate any model as testbeds? If so, can we do so based on their results against independent data?

② Methodology

5 Global Ocean Biogeochemical Models (GOBMs)*
7 stats/ML-based data products | 2001–2021

GOBMs in this study:	• FESOM-2.1-REcoM3	• CESM-ETHZ
	• MRI-ESM2-3	• NEMO4.2-PISCES (IPSL)
	• NorESM1-OCv1.2	

**Also repeated with 6 triple-member CMIP6 Earth System Models (ESMs)*

A. Models vs. independent data



1 Sample each model at independent data coverage

pCO_2 observational datasets: GLODAP, LDEO, & SOCAT



2 Compute statistical metrics of model skill against independent data

Metrics: $|Bias|$, uRMSE, correlation

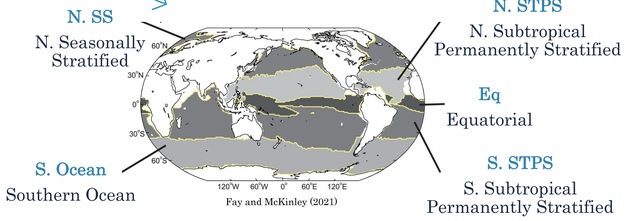
if lower metric = better skill,

$k = \frac{x - x_{min}}{x_{max} - x_{min}}$ (uRMSE and bias)

else, $k = \frac{x_{max} - x}{x_{max} - x_{min}}$ (correlation)



3 Rank models per "superbiome" and globally



Fay and McKinley (2021)

B. Testbed model choice



1 Define leave-one-out ensembles for every model



2 Compute the ensemble average of each annual metric

Collapsed dataset to time-series
Metrics: Bias, uRMSE, correlation



3 Repeat for all combinations of ensembles

Ensemble size: 1–5



4 Compute the spread, and where each leave-one-out ensemble sits in it

AI Statement

My contributions	AI contributions (Gemini Pro 3, Claude Opus 5)
Initial framework: "relative performance to best performer"; rank by sum of coefficients per biome	→ Suggested methodological improvements, e.g. ranking by L2-norm
LaTeX write-up of the improved methodology	→ Python code written from that LaTeX write-up + McKinley et al. (2026) code
Checked math; annotated code	Quick-plots for sanity checks
Interpreted and presented all results	← Made graphics for visuals

③ Results

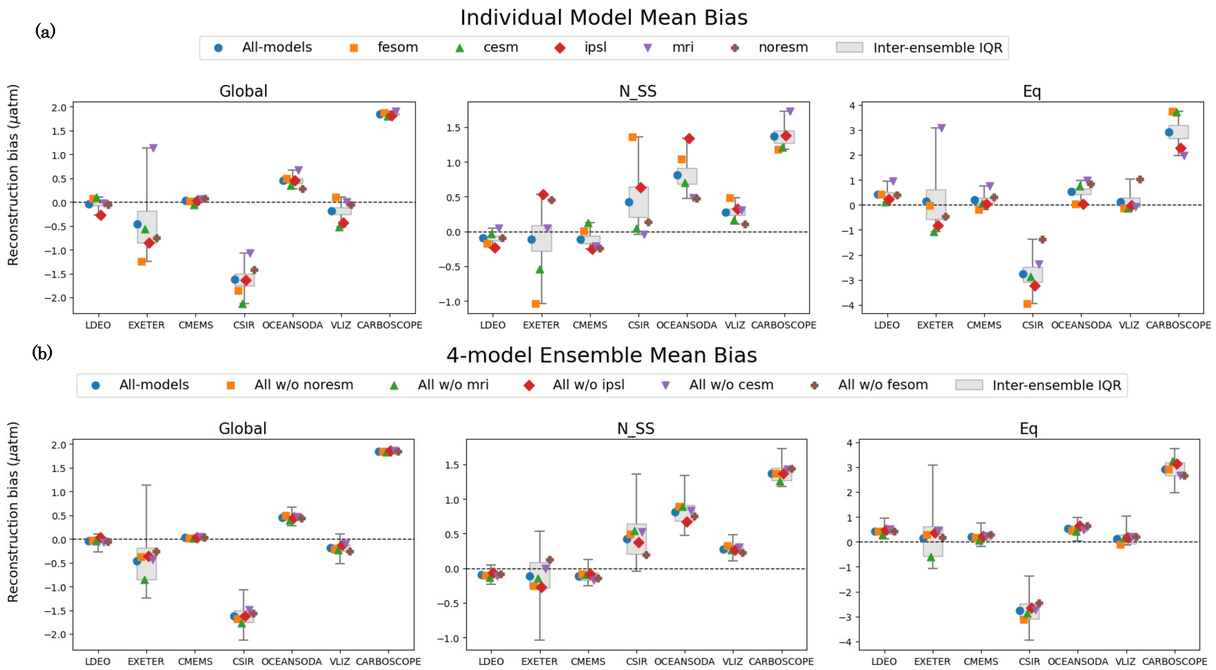


Fig. 1(a) Area-weighted 2001–2021 averaged annual global- and biome-mean bias of reconstructed surface pCO2. "All" testbed ensemble is composed of all 5 GOBMs. Grey boxes show the interquartile range (IQR) taken over all 31 possible ensemble combinations (1 to 5 members), with whiskers showing min-max. Testbed truths are masked with 100 and 1000 μatm cutoffs. **(b)** Averaged bias of leave-one-out ensembles by GOBM testbed choice.

Biome-mean metrics are *not* significantly impacted by testbed model choice!

Through ensemble-averaging, most regions and data products show narrower differences between testbed model choices.

Generally, including more members minimizes impacts from testbed model choice.

Outlier: including the MRI testbed in an ensemble creates visible impacts in numerous reconstruction methods, but only in some biomes (Eq & S.Ocean)

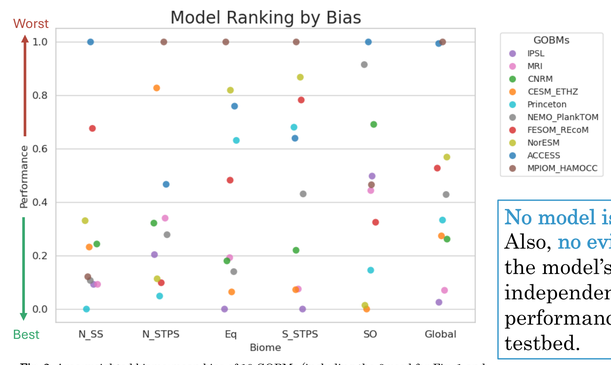


Fig. 2. Area-weighted biome-mean bias of 10 GOBMs (including the 6 used for Fig. 1 and 3) from Global Carbon Budget 2023. Ranked by absolute bias against independent pCO_2 data from SOCAT, LDEO, and GLODAP databases. Normalized scale to [0,1] where 0 is assigned to the model with the lowest absolute bias in the region, and 1 to the highest.

No model is globally "best". Also, no evidence suggests that the model's performance against independent data determines its performance when used as testbed.

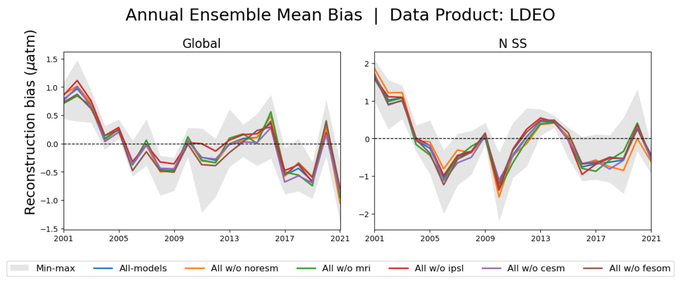


Fig. 3. Area-weighted annual mean bias time-series of the LDEO data product globally and in the N.SS superbiome. The shaded region shows the minimum-to-maximum spread of that year's annual mean across all 5 GOBMs. Testbed truths are masked with 100 and 1000 μatm cutoffs.

Annual time-series confirms that skill is driven by algorithmic choice of the reconstruction instead of testbed choice.

④ Conclusion

- Spread between testbed choices is large, but ensemble averaging reduces it
  - Including some specific testbeds can worsen reconstruction skill, but no choice of specific testbeds can improve skill
- Data product and biome choice, not testbed choice, drive reconstruction skill
  - Equatorial and Southern Ocean superbiomes are the most sensitive to testbed choice, and sensitivity also varies between data products

The project is repeated with triple-member CMIP6 ESMs (n = 6) → same conclusions

Is there evidence that we should *eliminate* any model as a testbed?

NO!

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